

SYSTEM REQUIREMENTS FEASIBILITY REPORT

E-WASTE BATTERY EXTRACTION SYSTEM **SYSTEMS & SIMULATIONS TEAM**

OBJECTIVE

The purpose of this report is to evaluate the hardware requirements of the main software platforms being considered for the project, namely Unity, MATLAB, and NVIDIA Isaac Sim. This report also assesses whether typical student systems can support these tools and identifies alternative solutions such as Deakin lab resources if higher computational power is required.

SOFTWARE HARDWARE REQUIREMENTS SUMMARY

The following table provides a comparison of the minimum and recommended hardware requirements for each software platform: -

SOFTWARE	MIN REQ	RECOMMENDED	COMPLEXITY
UNITY	8GB RAM, DXD10 COMPATIBLE GPU , SSD	16 GB RAM, dedicated GPU, SSD	EASY
MATLAB	8 GB RAM, 2-CORE CPU	16 GB RAM, SSD, GPU	EASY-MED
ISSAC SIM	32 GB RAM, RTX GPU, 16 GB VRAM, 50 GB SSD	High-end workstation (RTX 4080 or higher)	HIGH

Unity is the least demanding and is suitable for most development tasks including simulation and visualization. MATLAB is moderately demanding and suitable for modelling and analysis. Isaac Sim has very high requirements and is designed for advanced robotics simulation.

CURRENT SYSTEM CAPABILITY ASSESSMENT

At this stage, detailed hardware specifications of all team members are not available. However, based on typical student systems and my own device (Intel Core i7 processor with standard laptop configuration), a general assessment can be made.

Most student laptops typically have:

- 8–16 GB RAM
- Integrated or mid-range GPU
- Limited VRAM

Based on this:

- **Unity** is likely to run smoothly on most systems.
- **MATLAB** is also expected to run effectively for modelling and analysis tasks.
- **Isaac Sim**, however, is unlikely to run on standard laptops due to its requirement for high RAM and RTX-class GPUs with large VRAM.

This indicates that while basic simulation and modelling tools are accessible, high-end robotics simulation may not be feasible on personal systems.

RISKS AND LIMITATIONS

There are several risks associated with the use of these software tools:

- Isaac Sim requires high-end GPU hardware (RTX series with RT cores and high VRAM)
- Standard laptops may face performance issues or crashes
- Large storage requirements (up to 50 GB) may limit installation
- Limited ability to perform high-fidelity simulation locally
- Dependency on external computing resources for advanced simulation

These limitations may impact development speed and integration if not planned properly.

ALTERNATIVE SOLUTIONS AND RESOURCES

If current systems are insufficient, several alternative solutions are available:

- Use **Deakin RIoT Lab** for robotics development and testing
- Access **Apps and Desktops Anywhere** for MATLAB and other software
- Utilize university lab computers with better GPU capabilities
- Share high-performance systems within the team
- Use **Unity as a lightweight alternative** for simulation if Isaac Sim is not feasible

It is recommended to confirm availability of high-performance GPU systems with lab staff or supervisors before planning Isaac Sim usage.

FINAL RECOMMENDATION

Based on the analysis, Unity and MATLAB are the most feasible tools for initial development, as they are compatible with standard systems and support simulation, modelling, and system design effectively.

Isaac Sim, while powerful for robotics simulation, should only be used if access to high-performance GPU-enabled systems is confirmed through Deakin labs or other shared resources.

This approach will ensure smooth progress in early development stages while minimising hardware-related risks.

REFERENCES:-

- [Unity - Manual: Unity 6.3 User Manual](#)
- [What Is Isaac Sim? — Isaac Sim Documentation](#)